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Video Quality Prediction and Classification using XGBoost under Variable Encoding and Network Conditions

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ABSTRACT

This study presents a machine learning-based framework for video quality assessment (VQA). This framework enables the mapping of objective metric outputs, such as Structural Similarity Index Measure (SSIM) and Video Multimethod Assessment Fusion (VMAF), to subjective Mean Opinion Score (MOS). The absence of a standardized conversion from objective VQA methods to MOS poses challenges for consistency and comparability across different evaluation methods. The proposed mapping function delivers real-time estimation of user-perceived video quality, offering an effective alternative to traditional subjective tests, which are time-consuming, costly, and unsuitable for online services such as IPTV or streaming platforms. All constructed models were trained using the XGBoost algorithm. The dataset comprises over 700 distorted video sequences, spanning the two most prevalent codecs (H.264 and H.265), along with a wide range of resolutions, bitrates, and packet loss rates. Although real-world network traffic is affected by various impairments, including delay and jitter, packet loss is the primary factor driving video quality degradation. The SSIM-to-MOS and VMAF-to-MOS regression models achieved a Pearson correlation of 0.95 and an RMSE of 0.31. Additionally, we propose a reduced-reference classifier that estimates video quality using a limited set of video characteristics, including a frame (packet) loss rate derived from the original sequence. This approach enables the model to evaluate video quality without requiring a direct comparison between the original and the test sequence. The model achieves a weighted F1 score of 0.92 and is suitable for deployment in time-sensitive services such as live streaming and IPTV, even in environments where packet loss may occur. Benchmarking against a Back Propagation Neural Network (BPNN) and state-of-the-art methods confirmed the superiority of the described approach. All source codes are publicly available to support reproducibility and further research in Quality of Experience (QoE) assessment.

Introduction

In the realm of digital video distribution, optimizing perceived quality while minimizing bandwidth consumption is paramount. The development of efficient video codecs aims to achieve this balance. Striking an optimal trade-off between bitrate and visual quality is essential not only for reducing transmission costs and storage requirements, but also for ensuring a satisfactory user experience across diverse network conditions and playback devices. This balance is particularly critical in applications such as IPTV (Internet Protocol Television) services and Video on Demand (VoD) platforms, where network resources may be limited or frequently highly utilized. In contrast to Voice over Internet Protocol (VoIP), where network parameters such as low delay and jitter are essential due to the real-time and interactive nature of voice communication, video streaming services rely on client-side buffering, which effectively mitigates short-term variations in packet arrival. Consequently, packet loss and sustained bandwidth availability represent the primary determinants of perceived quality. A wide range of methods for estimating the perceptual quality of videos or images has been proposed in the literature. These methods can be classified into three main categories: full-reference (FR), reduced-reference (RR) and no-reference (NR). In RR methods, instead of providing the full original image or video (FR), only specific characteristic information (e.g. spectral coefficients or spatial-temporal analysis) is used to assess visual quality. In this work, we proposed both the FR and the RR metric. For the RR metric, we employed frame/packet loss as side information from the reference, representing the difference between the expected frames (from the source) and those actually received.

The following two FR objective metrics, SSIM and VMAF, are the most commonly used in scientific research and industry practice. The Structural Similarity Index (SSIM) is an objective metric for assessing the quality of an image or video relative to a reference source. It evaluates similarity across three perceptual components: luminance, contrast, and structure, and it yields a score between 0 and 1, with a score of 1 indicating that the test image/video is identical to the reference. Video Multimethod

Assessment Fusion (VMAF), developed by Netflix, is a modern video quality metric that improves upon traditional approaches. It leverages machine learning, resulting in scores that align more closely with human perception. The output ranges from 0 to 100, with higher values indicating better quality. Compared to conventional metrics such as SSIM, VMAF provides a more accurate and comprehensive assessment of video quality. Although both objective metrics are widely used for video quality assessment, their correlation with subjective quality expressed by the mean opinion score (MOS) can be limited. On the other hand, subjective VQA, while providing a more accurate representation of user experience, is resource-intensive, requiring substantial time and a large pool of human assessors for statistically significant results. Translating objective quality metrics scores into subjectively perceived quality results remains a fundamental challenge in VQA. While objective metrics quantify signal-level distortions, they fail to fully capture the complexity of the human visual system (HVS). Factors such as scene content, motion, temporal artifacts, and visual masking significantly affect perceived quality, often in ways not addressed by mathematical formulations. As a result, two videos with similar objective scores can produce markedly different subjective perceptions, leading to weak or inconsistent correlations with the MOS rating.

In particular, there is no official or standardized mapping function for converting SSIM or VMAF scores (each operating on a different scale) directly into MOS values, which limits their applicability in systems requiring accurate Quality of Experience (QoE) estimation. In the context of video streaming, QoE is a measure of a user's overall satisfaction with a service. For video service providers such as IPTV (fixed broadcast schedule similar to traditional TV) and VoD/OTT (Over The Top) platforms (allow viewers to choose and watch content at any time), accurately interpreting objective quality scores is crucial for maintaining user satisfaction and optimizing network resources. These metrics enable real-time monitoring and decision-making in adaptive streaming, load balancing and service-level assurance. However, misinterpretation can result in suboptimal quality adjustments, leading to a degraded user experience. Similarly, objective metrics play a critical role in video quality evaluation for state regulatory bodies. Regulators require reliable tools to assess customer complaints and verify the quality of service delivered by providers. Inaccurate conclusions from misinterpreting calculated scores can jeopardize consumer protection. This research addresses the challenge of accurately predicting subjective MOS scores based on objective quality metrics. To reflect real-world conditions, we constructed a dataset representing the most common video encoding profiles, covering a range of content types and compression settings. Video sequences were encoded using H.264 and H.265 standards at various bitrates and resolutions, allowing us to evaluate the SSIM, VMAF, and machine learning algorithms for their ability to estimate subjective quality. The main contributions of this paper are as follows:

- **XGBoost-based video quality prediction:** We developed a novel method using the XGBoost algorithm to estimate the video quality perceived by users from objective metrics (SSIM and VMAF), assigned to the MOS scale. To align SSIM more closely with VMAF (generally VMAF correlates slightly better with MOS than SSIM), the proposed mapping function applies a metadata-based approach that improves its correlation.
- **Reduced-reference classification model:** Our Reduced-Reference (RR) model evaluates the quality of a video by analyzing a limited set of features extracted from the source, removing the need for a full reference for calculation. Despite its reduced input complexity, it achieved a weighted F1 score of 0.92, demonstrating high effectiveness in real-world IPTV/VoD scenarios.
- **Feature importance analysis:** We identified key video parameters that significantly influence the subjective nature of perceived quality. SHAP (SHapley Additive exPlanations) values and Principal Component Analysis (PCA) were used to analyze and rank the importance of features.
- **Robust and reproducible implementation:** All implementation code is publicly available on GitHub and Zenodo. To ensure generalization, several regularization and validation techniques were applied to prevent overfitting during training.

The rest of the paper is organized as follows. The section on Related Work reviews selected important works in this research topic. The Methodology section presents the details of our XGBoost-based models. The Results section provides performance analyses and compares our models with state-of-the-art methods and relevant standards. The Discussion section outlines the benefits and limitations of the proposed approach. Finally, the Conclusion summarises the findings and discusses plans for future work.

Related work

As this research domain is very interesting and still being explored, many studies have been conducted in this field. Paper¹ has introduced an open-source framework for Quality of Experience (QoE) assessment in multimedia networks that relies solely on network-level parameters such as delay, jitter, and packet loss. Unlike traditional ITU-T P.1203 implementations, this approach eliminates the need for video-specific data while still achieving approximately more than 90% of the prediction

accuracy for Mean Opinion Score (MOS). The framework leverages machine learning, specifically random forest - the same approach we used, to model the relationship between network conditions and perceived quality, integrating real-time data collection and prediction in accordance with international standards. This reduced-parameter methodology significantly lowers computational demands and data requirements, offering a scalable and efficient solution for real-time QoE evaluation in diverse multimedia services. Study² addresses the growing dominance of video streaming services by introducing a continuous Quality of Experience (QoE) evaluation method grounded in Quality of Service (QoS) parameters. The approach integrates both subjective and objective assessment techniques to accurately estimate user QoE. To further enhance prediction accuracy, the authors develop a QoE-Ensemble MLP model that leverages ensemble learning with multilayer perceptrons. In addition, a low-complexity bandwidth allocation algorithm is proposed, enabling service providers to reduce network resource waste while maintaining user-perceived quality. Experimental results demonstrate the superior performance of the proposed evaluation framework and highlight the strong relationship between QoS metrics and QoE.

A further contribution seeks to overcome the limitations of traditional objective video quality assessment methods by applying machine learning techniques. In paper³, the authors propose a linear regression model trained on a set of objective quality metrics, enabling more accurate prediction of perceived video quality compared to individual metrics alone. The model demonstrates strong correlation with subjective user evaluations while maintaining a good statistical fit to empirical data. This approach extends classical objective methods by improving both efficiency and accuracy, offering a cost-effective solution for estimating user-perceived video quality. Paper⁴ suggests a user-centric QoS to QoE mapping model using back propagation neural networks (BPNN), integrating network parameters and Mean Opinion Score. It applies a combination of network optimization algorithms and enhances them based on the specific characteristics of the QoS/QoE relationship. The WRprop algorithm is introduced as an effective solution for this optimization. Work⁵ presents DeSVQ, a deep learning-based QoE prediction framework that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. The CNN extracts frame-level features from distorted videos, while the LSTM captures temporal dependencies from both visual and numerical features. Outputs from both stages are linearly combined and processed by decision trees for final QoE prediction. In paper⁶ authors propose a methodology that combines multiple objective quality metrics through small-scale subjective testing to enable scalable application-specific VQA. Pseudo-random sampling is used to select test content across resolution and quality ranges, guided by existing metrics. The subjective test results are then processed using support vector regression to find the optimal fusion of metrics for accurate quality prediction. Article⁷ introduces a novel Serial Dependence Modeling (SDM) framework for full-reference VQA, inspired by findings in neuroscience on serially dependent perception (SDP) in the human visual system. The research suggests that the HVS favors stable and predictable degradations over abrupt, unpredictable disturbances. The proposed SDM framework leverages this insight to improve the accuracy of video quality assessments. Paper⁸ proposes MO-QoE, a framework for video QoE prediction using Multi-Feature Fusion (MFF) and Optimized Learning Models (OLMs), which are tailored neural networks. The model uses Adaptive Moment Estimation and Batch Gradient Descent to optimally adjust weights and biases. By fusing multiple features, the MFF scheme avoids distortion-specific bias, resulting in a robust non-linear regression score for accurate QoE prediction. In work⁹ authors suggest CGRU-QoE, a QoE prediction model that combines CNN for extracting global features, GRU (a variant of LSTM) for capturing contextual information, and an attention mechanism. A new input factor—bitrate—is also incorporated to enhance prediction.

Paper¹⁰ proposes simple parametric models for predicting visual quality scores across various devices in multiscreen systems. The models use video objective metrics (PSNR, SSIM, VIF, VMAF) along with viewing setup parameters such as display size, resolution, and viewing distance. Validation on datasets from TVs, Ultra High Definition (UHD) TVs, and mobile devices confirms the models' effectiveness across different environments. Article¹¹ investigates the impact of Short-Term Memory (STM) on Quality of Experience through subjective experiments. Results reveal that users' real-time QoE is affected by prior viewing duration and expectations formed by STM. Analytical models are proposed to describe the relationship between intrinsic video quality, user expectation, and real-time QoE. Study¹² investigated machine learning-based video QoE estimation across different networks using a large dataset of 24,000 YouTube sessions collected in Germany and Croatia under controlled network conditions. They trained network-specific, general, and cross-tested models for multiple KPIs including MOS, resolution, stalling, initial delay, and bitrate. The results showed that network-specific and general models achieved similar performance, with initial delay classification reaching up to 99% accuracy, bitrate classification around 71–79% accuracy, MOS classification 65–83%, and resolution classification 56–62%. In contrast, cross-network testing significantly degraded accuracy, with MOS dropping to 35% and resolution to 36% when models trained on one dataset were applied to the other. These findings demonstrate that while general models can be reliably applied across multiple environments, cross-network transferability remains a challenge, requiring more advanced approaches such as drift handling or transfer learning for robust QoE estimation. Article¹³ proposes a machine learning method for overall video quality prediction using segment-level features in streaming sessions. Two feature sets are explored—one including segment quality, content characteristics, stalling duration, and padding; the other using bitstream-level parameters instead of content characteristics. These features are processed by an LSTM network to capture temporal relationships, with overall quality predicted through a linear regression layer. Another

contribution¹⁴ addresses the challenge of QoE monitoring in the context of end-to-end encrypted video traffic, where traditional deep packet inspection approaches lose effectiveness. The authors propose a machine learning-based framework leveraging a random forest classifier to infer QoE factors such as startup delay and stalling directly from network- and transport-layer features. Evaluated on YouTube traffic over HTTPS and QUIC, the approach achieves 91.1% classification accuracy for HTTPS and 87.3% for QUIC, enabling real-time detection of impairments without relying on payload inspection. This demonstrates the potential of lightweight ML models for accurate QoE inference under encrypted streaming conditions. Paper¹⁵ introduces a novel, low-complexity, no-reference (NR) VQA method. The proposed transformer-based architecture uses two specialised branches, one focusing on spatial distortion and the other on capturing spatiotemporal features. The model's performance in terms of the Pearson correlation coefficient on cross-database was below 0.8. This lower correlation is due to the fact that NR methods do not require any information from a reference. Work¹⁶ proposes a framework for monitoring Quality of Experience (QoE) in encrypted video streaming directly within the network, overcoming the limitations of deep packet inspection. The approach relies on network- and transport-layer features combined with machine learning models to infer QoE indicators such as startup delay and stalling events without requiring access to video payload. Evaluation on encrypted YouTube traffic demonstrates the effectiveness of the framework, achieving 93.9% accuracy for startup delay classification and 91.8% for stalling event detection. These results confirm that in-network QoE monitoring of encrypted video can be both accurate and efficient, offering network providers a practical tool for real-time service management. The AVQBits framework¹⁷ introduces a versatile bitstream-based video quality model applicable across diverse contexts such as service monitoring, video encoding evaluation, gaming QoE, and omnidirectional video quality. The model builds on the ITU-T P.1204.3 standard and is presented in four variants with varying input requirements. Trained on the AVT-PNATS-UHD-1 dataset and validated with AVT-VQDB-UHD-1, all models achieve strong performance, with Pearson correlations of 0.890 (Mode 0), 0.901 (Mode 1) and 0.928 (Hybrid Mode 0) against subjective user ratings. Evaluations further show that AVQBits generalizes well to emerging use cases including 360° video, high frame rate content (up to 120 fps), and gaming, where certain variants outperform even full-reference pixel-based approaches. Additionally, the paper introduces a long-term integration model based on ITU-T P.1203.3, capable of estimating overall audiovisual QoE for streaming sessions lasting from 30 seconds to 5 minutes.

A number of standardisation initiatives have aimed to address the challenge of estimating video quality in a way that can substitute subjective judgement. The ITU-T P.1203 standard¹⁸ provides a model for estimating audiovisual Quality of Experience in the context of HTTP Adaptive Streaming (HAS). It relies on a range of parameters such as bitrate, resolution, framerate, stalling events, and device or viewing conditions, to predict perceived quality. However, P.1203 is not designed to support UHD content or H.265 (HEVC) codec, which limits its applicability in modern streaming environments. To overcome these limitations, ITU-T P.1204¹⁹ extends video quality assessment to include UHD resolution and H.265 codec. It introduces a suite of model types tailored to varying application needs. Specifically, P.1204.4 is a reduced-reference (RR) model that leverages both original and processed video data to estimate perceived video quality. However, neither model can be deployed in a lossy environment. Paper²⁰ proposes a novel video quality prediction model for IPTV using unsupervised learning with self-organizing maps (SOM), aiming to bridge objective QoS metrics and subjective QoE evaluation. Unlike traditional regression or supervised models, SOM avoids overfitting, requires fewer training data, and provides intuitive clustering of video quality levels. A dataset of 432 distorted video sequences (H.264/H.265 encoded, various bitrates, packet loss rates, and content types) was created and evaluated using both SSIM (objective) and MOS via ITU-recommended ACR subjective testing. Feature selection with U-matrix visualization identified bitrate, packet loss, SSIM, and content type as the most influential inputs. The SOM classifier grouped inputs into three quality classes (green = high, orange = tolerable, red = poor), achieving a classification rate of 86% on a 10×10 topology and an AUC of 0.927, with a macro F1 score of 0.836.

Paper²¹ introduces DeepQoE, an end-to-end deep learning framework for generalized QoE representation learning to overcome limitations of existing QoE prediction models, such as dataset dependency, lack of configurability, and poor scalability with limited data. The approach leverages word embeddings and 3D CNN (C3D) features, which are fused and fed into a neural network to generate a unified representation usable for both classification and regression tasks. Evaluations on three datasets demonstrate substantial improvements: for small datasets like WHU-MVQoE2016 and the LIVE-Netflix Video Database, DeepQoE boosts performance of baseline machine learning models by 35.71%–44.82%, while on the large-scale VideoSet dataset, DeepQoE achieves 90.94% accuracy, significantly outperforming the best baseline method at 82.84%. Beyond predictive accuracy, the framework shows strong adaptability across diverse datasets and modalities, and its application to an adaptive bitrate (ABR) streaming system validates its practicality for real-world multimedia services. The software is released open source to support further QoE research.

In light of the rapidly evolving and dynamic nature of the research field, there was a strong focus on incorporating the most current and relevant studies (covering especially FR and RR metrics) to ensure that the work reflects the latest findings, even if some of these studies have received relatively few citations.

Methodology

The picture quality is influenced by several factors such as resolution, bitrate, or type of codec. As is well-known, this quality can be assessed through objective metrics (such as PSNR²², SSIM²³, and VMAF²⁴) or by subjective methods such as Absolute Category Rating (ACR) or Degradation Category Rating (DCR). Laboratory test conditions including resolution, viewing distance, ambient lighting, and test subject selection are detailed in²⁵, while²⁶ describes the mentioned subjective video quality assessment methods. Since subjective evaluation is time-consuming, it is essential to develop models that translate the score of objective methods into subjective results. Although some models do exist, this challenge remains highly relevant, and many research groups are actively working on it, as highlighted in the Related Work section.

Currently, the most commonly used video compression standard (codec) presented in a wide range of video streaming platforms (OTT) and IPTV is MPEG-4 H.264/AVC²⁷. Despite being released 20 years ago, based on²⁸, more than 80% of all available video content is still encoded also in this codec. Its successor, the H.265/HEVC codec²⁹, has suffered from the excessive royalties required to use it, which has slowed down its adoption. To date, almost half of the total video content is available in this video format. OTT and IPTV bitrates range typically between 5 and 15 Mbps. Premium streaming services offer even higher bitrates that can exceed 20 Mbps, however, this option often requires additional fees, and the customer must have high-speed Internet access. Since HEVC provides comparable picture quality at roughly half the bitrate³⁰ of its predecessor, a video bitrate of 15 Mbps should be adequate for the next few years — as widespread adoption of HEVC codec is expected. At present, High-Definition (HD) resolution represents the minimum requirement for IPTV and OTT streaming services, with FullHD and Ultra-High-Definition (UHD) formats progressively becoming the dominant standards. Therefore, for our dataset, we have used the following resolutions: HD (1280 x 720), Full HD (1920 x 1080), and Ultra HD/4K (3840 x 2160). The scene characteristics can be defined using two key indicators: Temporal Information (object motion) and Spatial Information (complexity), hereby referred to as TI and SI. The ITU-T P.910 recommendation advises assessing scene characteristics using these two indicators. It is important to ensure that the selected test scenes adequately cover both the spatial and temporal aspects to achieve sufficient diversity in the scenes being tested. The dataset was created using uncompressed UHD video sequences produced by the research group at Shanghai Jiao Tong University. The selected video sequences cover all quadrants of the SI-TI plot, ensuring that the full range of video content characteristics is represented in the analysis. All video sequences were encoded to video profiles that reflect the situation in popular OTT and IPTV services. For more detailed information about the preparation of the testing video sequences, please refer to our paper³¹. In that study, the subjective assessment was performed using MOS (the ACR method), while the objective metrics were SSIM and VMAF. In our previously published works^{20,32}, we have developed a mapping function that enables a classification/regression model to translate SSIM value into MOS. Models were based on Kohonen map and Back Propagation Neural Network (BPNN). Despite achieving promising results, we aimed to enhance our models by incorporating additional video scenes and integrating the VMAF metric—widely recognized in both the research community and industry for its effectiveness in video quality assessment. Instead of mentioned algorithms, in this study, we used decision trees with an implementation of the XGBoost (eXtreme Gradient Boosting) algorithm, but we also constructed BPNN models to use as a benchmark. A decision tree is a supervised machine learning algorithm used for classification and regression tasks. It works by splitting data into branches based on a set of features and their values, forming a tree-like structure that leads to a prediction. XGBoost is an optimized gradient boosting algorithm known for its speed, accuracy, and scalability. It is generally faster to train and more robust on small or imbalanced datasets than many other machine learning algorithms. Gradient boosting builds decision trees sequentially, with each new tree correcting the errors of the previous one by minimising the loss function using gradient descent as depicted in Figure 1. Although streaming platforms do not require precise MOS scores to assess video quality, they do need to know whether the quality remains acceptable. Codec developers, on the other hand, must verify their solutions against existing ones and therefore need to accurately interpret objective quality metrics in terms of subjective user perception. Therefore, we decided to evaluate XGBoost for the two types of task: regression and classification.

To construct our models, we used a dataset containing six different scenes (various of spatial and temporal information). The whole procedure of encoding the test sequences is described in our mentioned paper³¹. Each video sequence is defined by its resolution, bitrate, scene type, codec type (H.264/AVC or H.265/HEVC), and packet loss degradation ranging from 0 to 1%. The dataset is presented in tabular form and includes results from the two most widely used objective metrics, SSIM and VMAF, available for each video sequence. The expected (target) outputs, which reflect users' opinions for each test sequence, are represented as MOS ratings obtained from subjective ACR test sessions. These target values are crucial for improving the accuracy of a machine learning model. Comparing the predicted values (model outputs) with the target values allowed us to evaluate model performance. All mentioned prediction models were implemented in a Python environment.

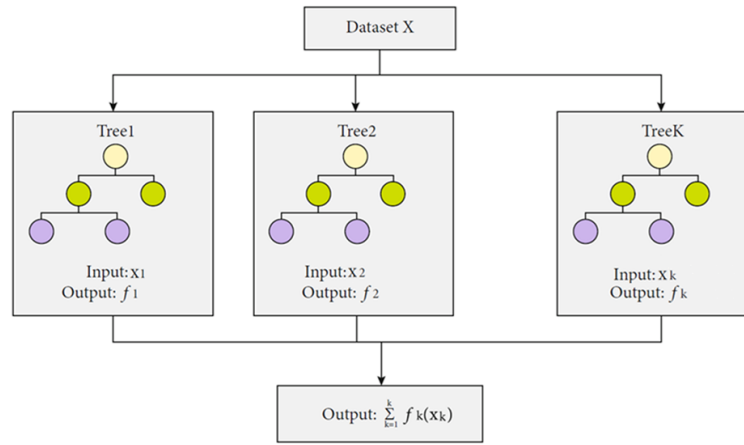


Figure 1. A general architecture of XGBoost.

Firstly, we aimed to determine whether it is necessary to use all planned model features (descriptors). We explored two approaches, SHAP (SHapley Additive exPlanations) and PCA (Principal Component Analysis). While PCA can only compute linear combinations of the original features, SHAP can identify nonlinear interactions between features. Figure 2 shows the importance and contribution of features to individual predictions for SSIM to MOS model. The SHAP method not only ranks the importance of each feature but also reveal the positive and negative influence of each feature on the prediction result.

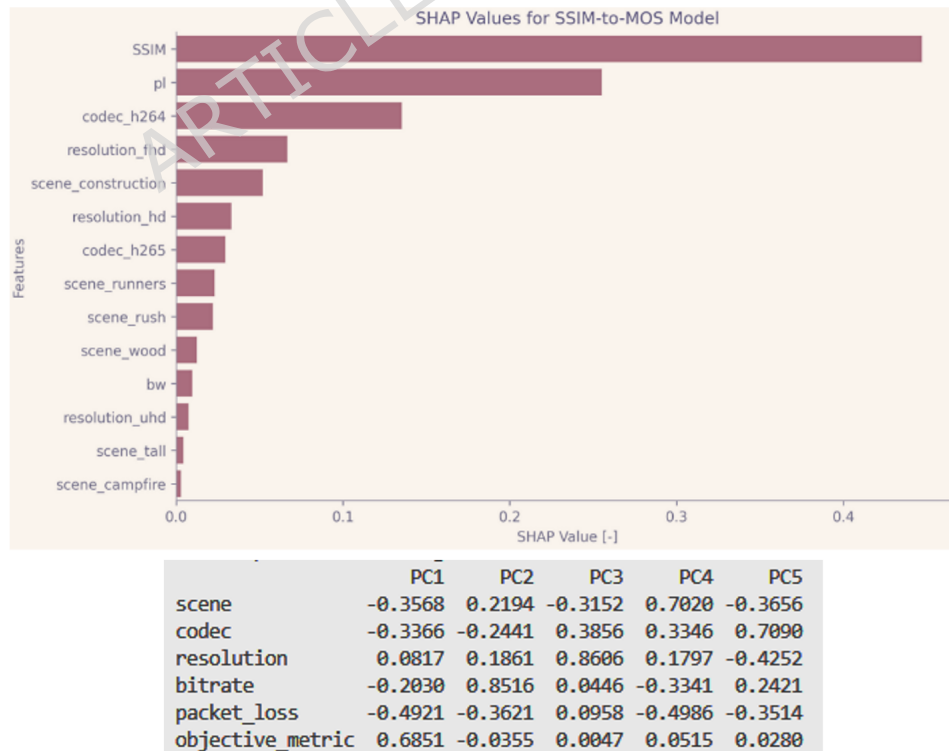


Figure 2. Analysis of importance for planned elements of input vector. SHAP (above) and PCA (below).

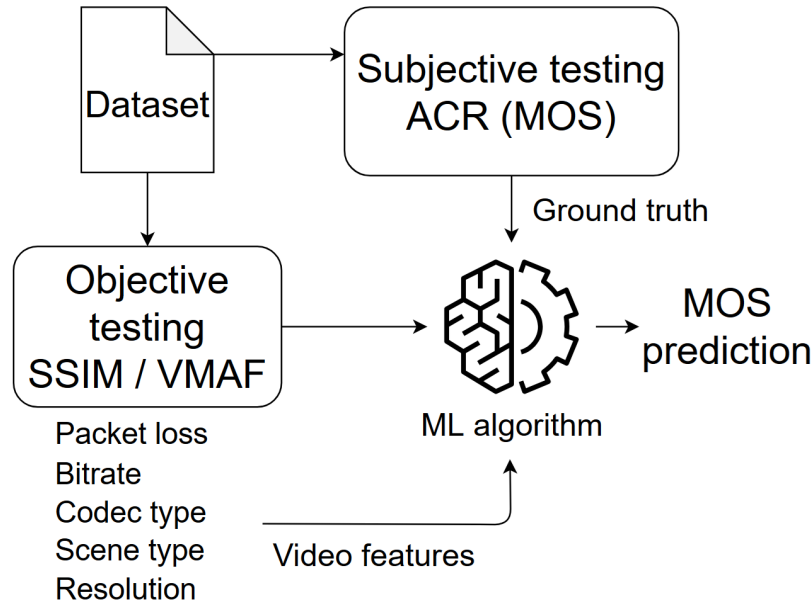


Figure 3. Diagram of the machine learning (ML) modelling process.

Both methods recognized codec and packet loss (pl) as parameters with the highest importance. The PCA analysis contains the variable weights for the five principal components (PC1 – PC5) which helps us understand which original variables contribute the most to each component. The negative sign indicates the direction of the relationship, not its impact. Based on this analysis, we decided to use all the descriptors mentioned. The proposed framework is depicted in Figure 3. Hyperparameter tuning refers to the process of finding the best set of parameters that control how the machine learning model learns and makes predictions. These parameters affect the model's complexity, learning process and ability to generalise. Hyperparameter optimisation was performed using grid search over a predefined set of parameters, including tree depth, learning rate, the number of estimators and regularisation terms. Each configuration was evaluated using stratified 3-fold cross-validation. The final model was selected based on the highest Pearson correlation coefficient/F1 score, with root mean square error (RMSE) or accuracy used as a secondary criterion. Below is a list of the selected parameters for XGBoost. The complete source code is freely available on Zenodo repository³³ and GitHub³⁴.

- **subsample** [0.4, 0.8]: Fraction (in percent) of training data used per tree. Each tree sees a different set of training samples. It helps prevent overfitting.
- **colsample_bytree** [0.5]: Percentage of features used per tree. Each tree uses only 50% of the feature set. Adds randomness and reduces overfitting.
- **max_depth** [3, 4]: Sets the maximum depth of a decision tree. A higher depth can produce a more complex model, which may overfit. Smaller values are typically used for small datasets (like ours).
- **learning_rate** [0.01, 0.05, 0.1]: Scales the contribution of each tree. Three values were tested.
- **n_estimators** [50, 100, 200]: Number of boosting rounds, i.e., the total number of decision trees the XGBoost model will train and add to the ensemble. Three values were tested.
- **alpha** [0.01, 0.03] (L1 regularization) and **lambda** [2, 3] (L2 regularization): Regularization parameters to control model complexity and prevent overfitting.

During the training process, we used a stratified K-fold split using the target variable (MOS values). This approach is commonly used for cross-validation, especially when dealing with classification tasks involving unbalanced class distributions. As shown in Figure 4, the dataset contains a small number of test sequences that have been evaluated with a MOS value of at least 4. The dataset was split into three folds, each containing a train and test subset in a 2:1 ratio. However, the samples within each fold were shuffled. This ensured that both the training and testing sets in each fold contained a proportionate distribution of MOS values. The training set comprises 563 samples per fold, yielding a total of 1,689 samples across the three folds. The remaining 20% of the samples (from 704) were used for testing only and were not used for training or tuning

of hyperparameters. The final performance metrics were computed exclusively on the test set. During cross-validation, the model was trained using the training subsets from two of the folds, while the remaining fold was used entirely as a test set for evaluation purposes. In the next iteration, a different fold was used as the test set and the remaining folds were used for training. This process was repeated until every fold had been used once for testing. Organising the data in this way enabled us to clearly track which samples were allocated for training and which for testing in each fold, as depicted in Figure 4, ensuring a fair and unbiased evaluation of the model's performance.

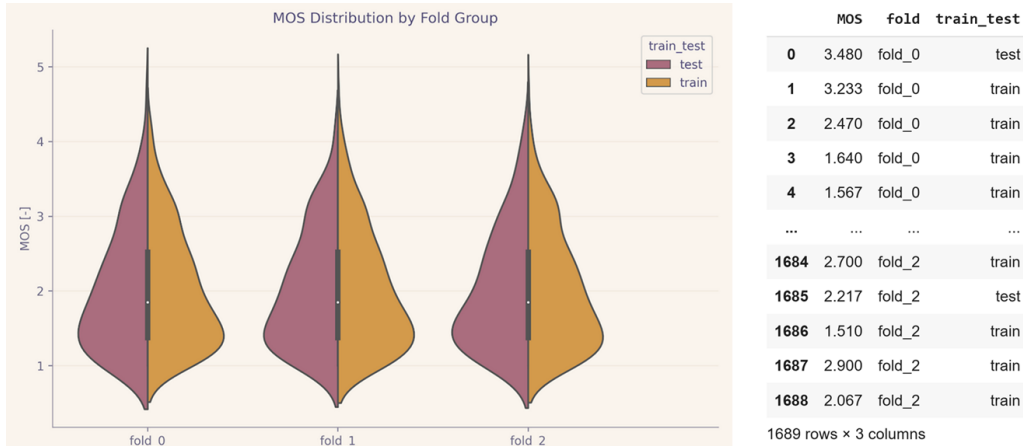


Figure 4. Stratified data split and distribution across each fold.

To conduct a comparative analysis of XGBoost with another widely recognised algorithm in the scientific community – Back Propagation Neural Network (BPNN), a SSIM-to-MOS model and a VMAF-to-SSIM model have additionally been developed, using the BPNN approach. In BPNN, hyperparameter tuning involves adjusting the number of hidden layers and neurons per layer, as well as selecting the activation function. The hidden layers transform the input vector into representations that are suitable for prediction or classification. This enables the network to capture non-linear relationships and approximate complex functions. The number of neurons in each layer decides which patterns can be extracted: too small a number may cause underfitting, while too large a number increases the risk of overfitting and raises the computational cost. Activation functions introduce the necessary nonlinearity, allowing the network to model complex mappings beyond linear combinations of inputs. The backpropagation algorithm computes the gradient of the loss function with respect to each weight, propagating errors backwards from the output layer to enable weight updates that minimise prediction error. Training usually involves gradient descent or its derivatives and may incorporate regularization techniques to enhance convergence, stability, and generalization. We used the Adam Optimizer, which automatically adjusts the learning rate for each weight. It keeps track of the average of past gradients and their squared averages to efficiently update the weights, helping the network to learn faster and more reliably. We tested a total of 53 configurations, varying the number of neurons per layer from 32 to 256 and the number of hidden layers from three to five. Additionally, we tested several activation functions, such as Sigmoid, ReLU, Swish and Tanh, on the topologies that were identified as the best. Overfitting was prevented using early stopping mechanisms (maximum of 200 epochs, it stops training when performance has not improved for a set number of epochs) and the dropout technique. Dropout is a regularization technique used in neural networks to prevent overfitting by randomly deactivating a subset of neurons during training, and in our case, a dropout probability of 5% was applied.

To assess the prediction veracity of the proposed models, we compared the predicted mean opinion scores (MOS) with the ground truth MOS values obtained from subjective quality assessment testing. Two performance metrics were used to measure the results for regression models: the Pearson correlation coefficient (PCC) and the Root Mean Squared Error (RMSE). The Pearson correlation coefficient measures the strength of the linear relationship between the predicted MOS $\hat{y}_i$, and the subjective MOS values y_i , and is defined as:

$$PCC = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}} \quad (1)$$

where N denotes the number of test samples, $\bar{y}$ is the mean of subjective MOS, and $\bar{\hat{y}}$ is the mean of predicted MOS. A higher PCC value (closer to 1) indicates a stronger agreement between subjective and predicted scores. RMSE is a metric that

evaluates the average magnitude of the prediction error, i.e. the difference between the predicted and subjective MOS values. It is computed as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (2)$$

Lower RMSE values indicate smaller deviations between the predicted and subjective ground truth scores, and therefore more accurate predictions.

The measurement of classification performance was performed using three evaluation metrics: Accuracy, Area Under the Curve (AUC), and F1 score. The weighted F1 score is a performance metric for multi-class classification problems. It accounts for class imbalance by averaging the F1 scores of individual classes, assigning weights that are proportional to their true sample counts. For each class i , the F1 score is computed from precision and recall as follows:

$$F1_i = 2 \cdot \frac{\text{precision}_i \cdot \text{recall}_i}{\text{precision}_i + \text{recall}_i}, \quad (3)$$

where

$$\text{precision}_i = \frac{TP_i}{TP_i + FP_i}, \quad \text{recall}_i = \frac{TP_i}{TP_i + FN_i} \quad (4)$$

where, TP_i , FP_i , and FN_i denote the number of true positives, false positives, and false negatives for class i , respectively. The weighted F1 score for N class dataset is obtained as follows:

$$\text{Weighted F1} = \frac{\sum_i^N (F1_i \cdot n_i)}{\sum_i^N n_i}, \quad (5)$$

where n_i represents the number of true samples of class i . This formulation ensures that classes with a larger sample size have a proportionally greater impact on the overall performance score, providing a more balanced evaluation in the presence of class imbalance.

Accuracy is defined as the proportion of correctly classified samples out of the total number of samples, and provides an overall measure of model performance. The AUC is a value between 0 and 1 that indicates the ability of a classifier to distinguish positive instances from negative ones, with values closer to 1.0 signifying stronger class separation. AUC represents the area under an ROC curve. The ROC (Receiver Operating Characteristic) curve visually represents the trade-off between correctly identifying positive cases and incorrectly identifying negative cases.

Results

Figure 5 illustrates the distribution of MOS values in relation to SSIM and VMAF metrics. While VMAF shows a more linear characteristic, most SSIM values fall within the 0.7–1 range. Since the MOS value is not always sufficient to describe individual picture quality, we also investigated the so-called Fair or Better (FoB) parameter. FoB is defined as the proportion of users assigning a score of 3 or higher to a testing video on the 5-point ACR scale. For our dataset, we found that a MOS value of around 2.6 indicates a situation in which at least half of the observers rated the testing sequence as 3 or higher, which is in line with previous published results.^{35,36}



Figure 5. MOS distribution and relationship between MOS and FoB.

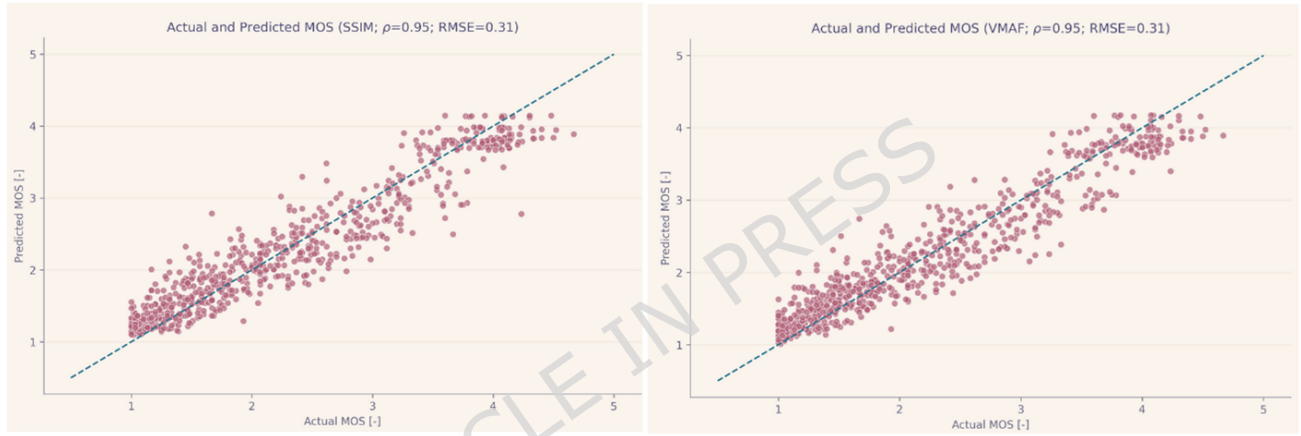


Figure 6. Quality of predictions evaluation for proposed XGBoost based model for SSIM to MOS (left) and VMAF to MOS (right).

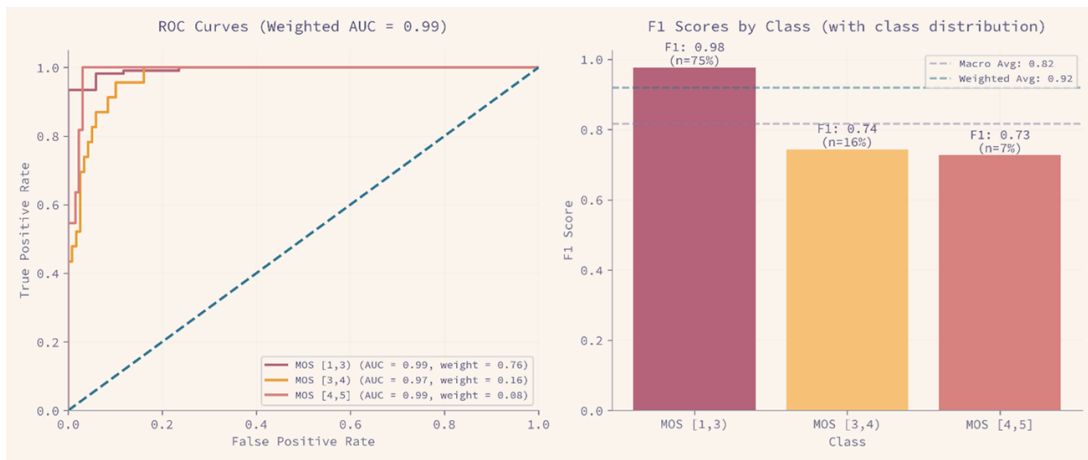


Figure 7. XGBoost based classification model.

Figure 6 depicts the Pearson Correlation Coefficient (ρ) and Root Mean Square Error (RMSE) for our proposed models. Gained RMSE of 0.31 on the 5-point MOS scale is considered sufficiently precise by us.

Decision trees and their XGBoost extension are also widely recognized as effective tools for classification tasks. In real-life situations, IPTV or OTT providers often find the exact MOS value unnecessary. What matters is a reliable classification of whether the delivered video stream meets an acceptable quality threshold. For this purpose, we proposed a XGBoost-based classification model that categorizes picture quality as good ($MOS \geq 4$), fair ($3 \leq MOS < 4$) or poor ($MOS < 3$). These three categories reflect the MOS verbal scale used in recommendation P.910. The results are depicted in Figure 7. The input vector did not contain any objective metrics. Predictions are calculated based solely on the video features mentioned above. The constructed model is highly effective at predicting poor streaming quality. However, due to the unbalanced dataset, the next two classes are represented by a smaller amount of data. Nevertheless, the weighted F1 score is above 0.9 (specifically, 0.92), indicating good predictive accuracy. As this model does not employ objective video quality descriptors but uses the packet loss rate (derived from the source frames information), it is classified as a Reduced-Reference (RR) metric. Unlike Full-Reference (FR) metrics, which compare a distorted video sequence directly to an original reference sequence to assess quality, RR metrics use only partial information from the reference sequence by extracting features or statistics (e.g. packet loss) to evaluate the quality of the video stream. While RR metrics deliver predictions more quickly and without needing a full reference sequence, they tend to have lower accuracy. Additionally, we also developed an RR regression model using XGBoost. As can be seen in Figure 8, the RMSE and correlation have dropped slightly compared to the proposed FR models (Figure 6).

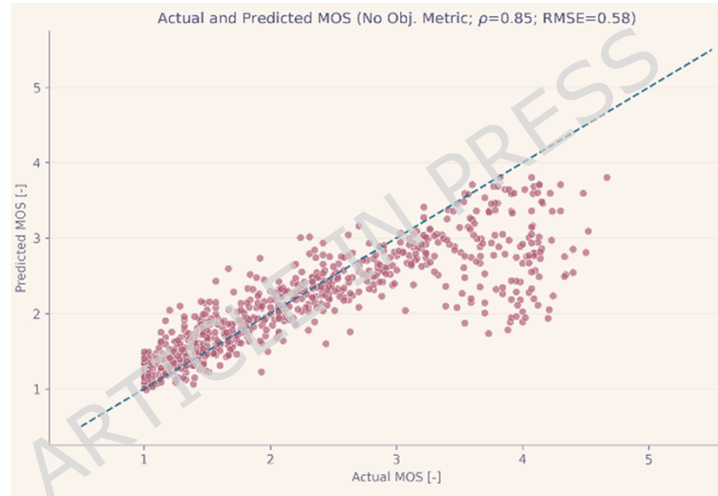


Figure 8. Proposed RR model for video quality estimation.

One of the objectives of this paper is to rigorously evaluate the predictive capabilities of XGBoost for video quality estimation. To achieve this, XGBoost was benchmarked against BPNN, a well-established model known for effectively capturing complex nonlinear patterns. This comparative analysis highlights the strengths and limitations of both methods, providing valuable insights into their suitability as mapping functions for converting objective metrics into MOS ratings.

The final settings for the FR models based on BPNN are as follows: The network topology (hidden layers) was 150-80-64 for SSIM and 264-128-65-32-16 for VMAF. The activation function was ReLU for SSIM and swish for VMAF. The Swish function was proposed by Google researchers in 2017 and can outperform ReLU in deep networks due to its non-monotonic and smooth characteristics³⁷ Results are shown in Figure 9.

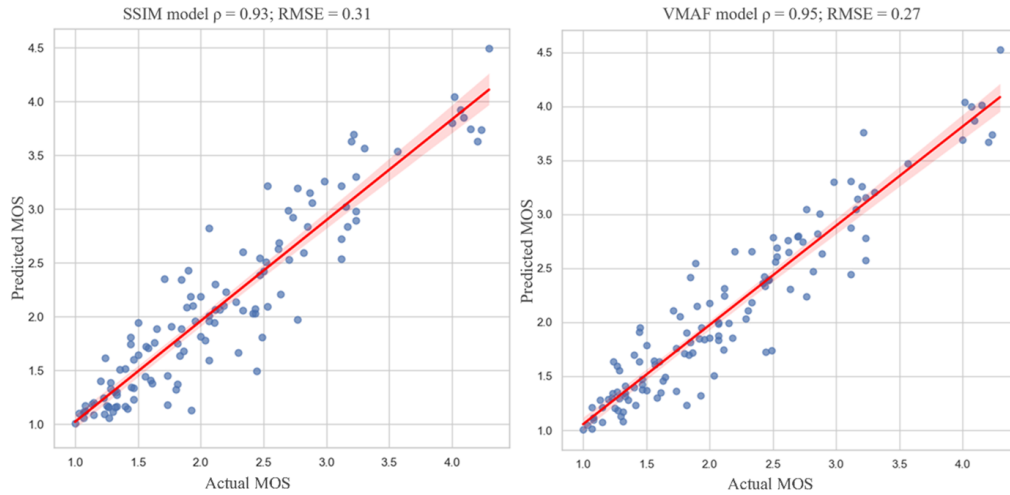


Figure 9. Quality of predictions evaluation for proposed BPNN based model for SSIM to MOS (left) and VMAF to MOS (right).

Based on the delivered results, we can conclude that XGBoost is a strong competitor to BPNN for this type of prediction task. For SSIM to MOS model decently overperformed BPNN and for VMAF to MOS model produced predictions with the same level of accuracy.

Discussion

The results presented in this study highlight the strong potential of XGBoost as a regression and classification tool for predicting video quality as perceived by end users. Our approach converts computed scores from objective quality metrics such as SSIM and VMAF, into subjective Mean Opinion Score. MOS is used to subjectively assess the perceived quality of visual content based on human judgment. Subjective tests are useful for capturing human perception and experience, which objective metrics alone cannot fully reflect. To achieve the best possible correlation with human ratings, we included additional information related to the video coding profile in the input vector, such as bitrate, resolution, and scene dynamics. Compared to the BPNN benchmark models, the XGBoost-based models demonstrated equivalent or superior performance, particularly in SSIM-to-MOS regression, achieving a slightly higher correlation coefficient ($\rho=0.95$ vs. $\rho=0.93$ for the BPNN model). This suggests that tree-based ensemble methods may be better suited to capturing complex relationships in small, imbalanced datasets. The integration of SHAP and PCA analyses provided additional insight into feature relevance. Both methods identified codec type, packet loss and scene type as the main factors influencing perceived quality. This confirms that network-related degradations and compression algorithms play a pivotal role in shaping the user experience. These results supported the decision to include all planned descriptors, as they meaningfully contributed to prediction accuracy.

In addition to evaluating accuracy, we also assessed computational efficiency. The training time for the XGBoost-based models remained under a few minutes for the entire dataset. In comparison, the BPNN benchmark took 5–6 times longer. These results further support the suitability of XGBoost for integration into real-time monitoring systems where latency and resource constraints are critical.

The innovative approach involving a classification model based on XGBoost, which does not rely on any Full-Reference metrics, represents a significant advancement in Reduced-Reference video quality estimation. Although it entails a slight trade-off in accuracy (as reflected in RMSE and correlation metric), it offers greater deployment flexibility and lower computational cost. IPTV providers, for example, do not need to know the exact MOS value — they only need to determine whether the delivered quality is sufficient. Notably, our proposed RR classifier achieved a high weighted F1 score of 0.92, highlighting its effectiveness in identifying unacceptable quality levels.

To demonstrate the effectiveness of our solutions, we assessed their predictive performance against state-of-the-art models. The comparison is shown in Tables 1, 2, 3, where each table represents particular model.

Table 1. Full reference models – regression

Reference	ML algorithm	ρ	RMSE	Model constraints
Chadha et al ⁶	SVR (SSIM to MOS)	0.79	1.17	no UHD, no H.265, no packet loss
	VMAF to MOS	0.91	0.86	
Liu et al ⁷	LSTM (SSIM to MOS)	0.92	N/A	no packet loss, no H.265
Ghosh et al ⁵	LSTM+XGBoost	0.90	0.35	
Ghosh et al ⁸	BPNN	0.94	0.43	H.265 not tested
Klink et al ³	Linear reg. (SSIM to MOS)	0.94	0.41	no UHD, no packet loss
	VMAF to MOS	0.95	0.38	
Zhang et al ⁹	LSTM (SSIM to MOS)	0.89 (SROCC)	0.54	no packet loss
Ying et al ¹¹	Linear weighted model (SSIM to MOS)	0.92	0.45	no UHD, unknown codec used
Barman et al ¹⁰	Linear model (SSIM to MOS)	N/A	0.44	
	VMAF to MOS	N/A	0.35	missing info about packet loss
Proposed models	XGBoost (SSIM to MOS)	0.95	0.31	
	XGBoost (VMAF to MOS)	0.95	0.31	
	BPNN (SSIM to MOS)	0.93	0.31	
	BPNN (VMAF to MOS)	0.95	0.27	

Note: SROCC - Spearman's Rank Correlation Coefficient; N/A - Not Available

Table 2. Reduced reference models – regression

Reference	ML algorithm	ρ	RMSE	Model constraints
ITU P.1203 ¹⁸	log-linear regression	0.89	0.33	no packet loss, no H.265, no UHD
ITU P.1204.4 ¹⁹	log-linear regression	0.88	0.44	
AVQBitsM1 Rao et al ¹⁷	linear regression	0.90	0.48	based on ITU-T P.1204.3, no packet loss
AVQBitsM0 Rao et al ¹⁷	linear regression	0.89	0.50	based on ITU-T P.1204.3, no packet loss
Proposed model	XGBoost	0.85	0.58	

Table 3. Reduced reference models – classification

Reference	ML algorithm	Accuracy	F1-score	Model constraints
Frnda et al ²⁰	Kohonen map	86%	0.84	
Tisa-Selma et al ¹⁴	Random Forest	91%	0.91	no packet loss, unknown codec used
Orsolice et al ¹²	Random Forest	83%	0.83	no packet loss, unknown codec used
Orsolice et al ¹⁶	Random Forest	84%	0.84	no packet loss, unknown codec used
Proposed model	XGBoost	89%	0.92*	

Note:* Weighted F1-Score

The proposed models demonstrate strong performance across both Full-Reference and Reduced-Reference metrics, rivaling or surpassing recent state-of-the-art approaches. In the full-reference domain, the XGBoost and BPNN-based models, whether using SSIM or VMAF as input features, achieve consistently high correlation coefficients ($\rho = 0.93$ – 0.95) and the lowest RMSE values (0.27–0.31) among all compared works. Although SSIM typically shows lower correlation with subjective MOS scores, the applied mapping function significantly improved its performance, achieving a correlation level comparable to VMAF. Furthermore, the application of linear regression approaches (e.g., Klink et al³ or Barman et al¹⁰) has been shown to generally yield higher errors (RMSE > 0.38), thus confirming the advantage of non-linear learners such as XGBoost in the modelling of complex quality dependencies.

In the reduced-reference regression models, the proposed XGBoost method achieves a correlation that is slightly below both ITU standardised models. It is important to note that the ITU models impose stricter limitations, such as the exclusion of packet loss, H.265, or UHD content. In contrast, our proposed model is less constrained. This renders it potentially more suitable

for deployment in diverse or evolving video delivery environments. The reduced-reference classification model proposed in this study proves to be highly competitive. It achieved an accuracy of 89% (correctly classified samples 630 from 704), outperforming several previous approaches based on different machine learning algorithms. The model demonstrates exceptional performance in classifying the MOS <3 class, achieving an accuracy of approximately 98.6% by correctly identifying 508 out of 515 samples. This high level of reliability in detecting low-quality content is a critical and successful outcome, as the model's primary purpose is to serve as an early warning component in monitoring systems. Unlike many prior models, which often omit codec information or avoid testing under packet loss conditions, the proposed model maintains strong performance despite its simplified design. This highlights its potential for efficient integration into real-time monitoring video systems, without relying on full-reference metrics.

While the dataset utilized in this study incorporates a wide range of encoding profiles, encompassing multiple resolutions (HD to UHD), bitrates, and two prevalent codecs (H.264 and H.265), the dataset also exhibits certain limitations that may impact the generalisability of the proposed models. Firstly, the data set contains only six unique video scenes, which restricts content diversity. Despite employing a stratified 3-fold cross-validation scheme and an independent hold-out test set, the generalisation capability of the proposed model remains constrained by the characteristics of the available dataset. Although the data set contains various video scenes with varying motion and visual complexity, all samples originate from the same data set, which may introduce content-related bias. However, to the best of our knowledge, there is currently no publicly available dataset that simultaneously combines packet loss impairments (up to 1%), video resolutions up to 4K, and subjective and objective quality annotations, including metrics such as ACR, VMAF and SSIM for both codecs. This limitation restricts the possibility of conducting cross-dataset validation or performing direct comparisons with alternative approaches under comparable conditions. Therefore, constructed models provide a solid baseline, showing how machine learning can map subjective ACR ratings to objective quality metrics and how these methods can be applied in practice. Although the classification model is highly effective at detecting significantly degraded quality (MOS <3), its accuracy at identifying moderate and higher quality levels is less consistent. Consequently, it is better suited for providing an early warning system than predicting precise differences in video quality. Secondly, the codec coverage is restricted to the two mentioned codecs, excluding more recent or increasingly adopted formats such as AV1 or VVC (H.266). Although these codecs currently possess a relatively minor market share in the context of streaming, it is anticipated that their adoption will undergo expansion, which may consequently give rise to the challenge of the long-term applicability of the model. Thirdly, the dataset incorporates a single framerate setting, which may influence perceived temporal smoothness – an important factor in video quality assessment. Finally, although the loss of packets was incorporated into the distortion factor, the range of the tests was limited to low levels (0.1%–1%). It is acknowledged that this may not fully capture conditions encountered in unstable or mobile networks. However, based on prior experimentation, packet loss exceeding 1% has been shown to result in a significant degradation of visual quality, causing the video to become unwatchable³². A further limitation of this study is the restricted comparison of learning algorithms. The main reason we did not test other models, such as convolutional networks or transformers, is that these methods either perform poorly with small, imbalanced datasets or have significantly higher training and inference costs (prediction speed). Our design objective was to balance predictive power with deployment efficiency, making tree-based ensembles the most practical choice.

Conclusion

In summary, this study validates the applicability of XGBoost as a scalable, interpretable, and efficient alternative to other machine learning models for both Full-Reference (FR) and Reduced-Reference (RR) video quality assessment. Its practical advantages, combined with competitive predictive performance, suggest that it is a viable candidate for deployment in real-world QoE monitoring systems. We developed models capable of mapping score from objective quality metrics into the subjective MOS scale. Notably, SSIM was modified to align its correlation with subjective scores to that of VMAF. Our RR regression model is slightly less accurate than ITU-standardised models, but it can be used in lossy transmission environments, such as IPTV services. Furthermore, the proposed RR classification model outperformed several published alternatives and offers straightforward integration into monitoring systems operated by IPTV or Video-on-Demand (VoD) providers. In future work, we aim to expand the dataset by incorporating recent codecs such as AV1 and VVC, and by extending the resolution range up to 8K (7680 x 4320), enabling more comprehensive modeling across diverse video delivery scenarios.

Data Availability Statement

The dataset employed in this study is publicly documented in Reference 31, which describes its characteristics and provides access guidance. The source code for the models presented in this study is freely available via the repositories listed in References 33 and 34.

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Author contributions statement

J.F.: Conceptualization, Methodology, Formal analysis, Resources, Validation, Software, Writing - Original Draft; J.R.: Software, Visualization, Data Curation, Investigation; M.U.: Resources, Conceptualization, Writing - Original Draft; J.B.: Data Curations, Writing - Review and Editing; M.V.: Formal analysis, Supervision; All authors reviewed the manuscript.

Additional information

Competing interests The author(s) declare no competing interests.